

Synergetic effect of graphene nanoplatelet, carbon fiber and coupling agent addition on the tribological, mechanical and thermal properties of polyamide 6,6 composites



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ABSTRACT

Polyamide 6,6 (PA6,6) is one of the commonly used engineering polymers and it is used for various applications. Moreover, tribological and mechanical performance of PA6,6 can be improved with the addition of fibers or particulates. However, properties of reinforced PA6,6 matrix composites are affected by many factors. Interfacial adhesion between the reinforcement and matrix material is one of these factors and there should be good interfacial adhesion to obtain good ultimate properties. In this study, it was aimed to improve tribological and mechanical properties of carbon fiber reinforced PA6,6 composites by means of improving the fiber-matrix interaction by using graphene nanoplatelet (GNP) and 1,4-phenylene-bis-oxazoline (PBO). Adhesive wear test, tensile test, dynamic mechanical, differential scanning calorimetry and scanning electron microscopy analyses were performed. Consequently, all the test results manifested that CF_0.5GNP_PBO coded composites exhibited improved tribological and mechanical properties among the all composites.

1. Introduction

Carbon fiber (CF) reinforced polymeric matrix composites are commonly used for many kinds of industries due to their outstanding properties such as good mechanical strength, low density and correspondingly weight saving property, good chemical and corrosion resistance etc. [1,2]. Moreover, there are still possibilities for the further improvement ultimate properties and extension of their usage area by considering some key factors. The most important one of these factors is the quality of interfacial adhesion between fiber and matrix, due to the interface, between fiber and polymeric matrix, acts as a bridge between these constituents and transfers the applied load from polymeric matrix to fiber. If the quality of interfacial adhesion between these constituents is good, a more efficient load transfer takes place with the fibers carry more load. Correspondingly, ultimate performance of composite improves [1]. However, improving the adhesion quality between CF and polymeric matrix is still a challenge because of the non-polar nature of the CF and inadequacy of functional groups on its surface and correspondingly its weak adhesion ability to polymeric matrix [2–4]. To date, there have been some methods developed by many researchers to improve the fiber-matrix adhesion quality. These methods can have classified as (1) fiber surface modification methods; (2) matrix

modification methods; and (3) development of hierarchical micro/nano composites [3]. Although fiber surface modification is a more efficient method when compared to matrix modification, it requires numerous chemical processes. Because of this reason, matrix modification method is more practicable and easier to apply [2].

As mentioned above, production of hierarchical micro/nano composites is another method to improve the adhesion quality between fiber and matrix. Particularly in recent studies, carbon base nano-size fillers such as graphene (GN) and carbon nanotubes (CNTs) have been widely used for this purpose [1–17]. In one of these studies, Chen et al. [10] placed a gradient interphase, reinforced by graphene sheets, into CFs and matrix, in order to improve the interfacial properties of CF reinforced epoxy matrix composites. Their results revealed that flexural strength and interlaminar shear strength properties of the composites are enhanced due to the formation of gradient interface layer and correspondingly improved interfacial properties. In another study, Zhang et al. [16] aimed to improve the interfacial adhesion in CF reinforced epoxy matrix composites by co-grafting CNT and polyhedral oligomeric silsesquioxanes onto CF surface. They found that surface functionality of the CFs was increased after modification process and consequently, a remarkable improvement was observed in the interfacial adhesion strength property of the new developed hybrid

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composites. Pathak et al. [9] investigated the effects of graphene oxide addition to CF reinforced epoxy matrix composites on the mechanical properties of hybrid composites. Their results revealed that mechanical properties such as flexural strength, flexural modulus and interlaminar shear strength were increased in the presence of graphene oxide due to the improved interfacial interaction between graphene oxide, epoxy matrix and CF.

On the other hand, there are some drawbacks for carbon base nano-size fillers to perform their function. While one of them is the difficulty of the homogenous dispersion of fillers in polymer matrix, another one is the high cost of manufacturing process of these fillers [18]. In this regard, platelet shaped graphene layers (can also be named as graphene nanoplatelets (GNPs)) exhibit more efficient reinforcement effect in some ways, when compared to some kind of carbon base nano-size fillers. In addition, GNPs has distinctive properties such as relatively low manufacturing cost when compared to graphene oxide and significantly high surface area when compared to CNT [18]. In addition to this, while mechanical performance of GNP is similar with CNT, its surface area is higher than it is. The reason of this case is while GNP possesses two-dimensional lattice structure, CNT possess one-dimensional lattice structure. Consequently, with high mechanical strength and high surface area, which enables better stress transfer between polymer and filler, GNP is a more preferable reinforcement material than CNT. However, platelet structure of GNP exhibits some disadvantages and the most important one of these disadvantages is the agglomeration problem. Because considerable extent of van der Waals bonds take place between GN layers and these interactions result in the form of GNP agglomerates. Hence, surface area of GNP decreases with increasing agglomerates and correspondingly reinforcement performance of it deteriorates [1,2,7,18–20].

Even if simultaneous usage of GNP and fiber overcomes the agglomeration problem somewhat, there is still a need for other ways to obtain efficient solution to this problem. One approach to overcome this problem may be the simultaneous usage of “matrix modification” method with the “production of hierarchical micro/nano composites” method. Matrix modification method is a process which polymeric matrix is functionalized by using chemical or physical method. By using this method, interaction between functionalized polymer and reinforcement material increases and correspondingly ultimate properties of composites improve. This polymer functionalization process can be performed in two different ways, physical and chemical. In chemical modification process, basic principle is grafting the reactive molecules onto polymer chains by fulfilling the required process conditions and thus, functionalizing the polymer [21]. In the matrix modification process, some polar coupling agents, which chemically affiliated with reinforcement surface, can be used in order to improve the interfacial interaction between matrix and reinforcement material. The coupling agent used in this method is a kind of chemical, whose mission is to form a bridge between matrix and reinforcement material. Accordingly, it is assumed that while one side of coupling agent reacts with surface of reinforcement material, other side of it reacts with polymeric matrix and improves interfacial interaction by acting as a bridge [22]. In our previous studies [23–25], efficiency of matrix modification method by using coupling agents on the mechanical and adhesive wear properties was proved for glass fiber and CF reinforced polyamide matrix composites.

As it has been mentioned already, there are many studies in the literature reporting the effect of carbon base nano-size filler addition on the properties of fiber-reinforced polymer matrix composites. While in some of these studies, nano-size fillers are directly added to manufacturing process of composites [2,4,19], in some of them, fibers are coated with nano-size fillers before composite manufacturing process [26–29]. In one of these studies, Yu et al. [4] investigated the effect of carbon base nano-size filler addition on the properties of CF reinforced epoxy matrix composites. Their results revealed that hybrid composites exhibited better performance than only CF included composites. In

another study, Yang et al. [2] prepared poly(arylene ether nitrile) (PEN), CF and GNP hybrid composites (PEN/CF, PEN/GNP and PEN/CF/GNP) and they investigated the possible synergetic reinforcing effects between these constituents. Their results showed that an apparent synergetic effect on the thermal and mechanical properties of composites was observed when compared to PEN/CF and PEN/GNP composites. Chen et al. [26] investigated the tribological properties of CNT grafted-CF reinforced epoxy matrix composites. They found that tribological properties of the CF reinforced epoxy composites significantly improved after grafting CNT onto the CF surface. Shen et al. [19] prepared the CF reinforced epoxy matrix composites and they added GNP to composites in order to obtain improved mechanical properties. As a result of their study, they observed a significant improvement in mechanical properties for GNP included composites. However, as a result of literature survey, none of these investigated studies focused on the properties of hierarchical composites which are prepared by using matrix modification method in the presence of a kind of coupling agent.

As mentioned in the literature survey above, there have been many studies in the literature, which investigated the efficiency of hierarchical reinforcement on the ultimate properties of composites. However, within our knowledge, so far there have not been any reported studies on the tribological, mechanical and thermal properties of CF reinforced polyamide 6,6 (PA6,6) matrix composites in the presence of GNP and 1,3-phenylene-bis-oxazoline (PBO) coupling agent. Therefore, this study aims to investigate how the tribological, mechanical and thermal properties of CF reinforced PA6,6 composites are affected by the simultaneously addition of GNP and coupling agent. In this research, tribological, mechanical, thermomechanical, thermal and morphological properties of composites were studied by adhesive wear test, tensile test, dynamic mechanical analysis (DMA), differential scanning calorimetry (DSC) and electron microscopy (SEM) analyses.

2. Materials and methods

2.1. Materials and composite preparation

PA6,6, purchased from Epsan (Turkey), was used as matrix material. As-received GNPs (average diameter of 1.5 μm , thickness of 3 nm, purity of 99.9% and surface area of 800 m^2/g) were used in this study and purchased from Nanografi (Turkey). Chopped CFs (average length of 6 mm) were kindly supplied from DowAksa (Turkey). Manufacturer indicated that these CFs were treated with a polyamide compatible sizing material. Coupling agent PBO was purchased from Evonik (Germany).

In this study, melt-mixing method was used in the composite preparation process. The reason for choosing this method is the convenience of it and its correspondence with polymer processing techniques used in industry. In this sense, melt mixing was performed by using laboratory type micro compounder with screw speed of 100 rpm and barrel temperature of 285 $^{\circ}\text{C}$. Immediately afterwards of melt-mixing process, test samples were produced by using injection molding method. For this purpose, laboratory type injection molding machine was used to sample preparation process, with an injection pressure of 10 bars, mold temperature of 80 $^{\circ}\text{C}$ and barrel temperature of 285 $^{\circ}\text{C}$. PA6,6 was pre-dried in a vacuum oven at 80 $^{\circ}\text{C}$ for 24 h Before the compounding process. While CF and PBO loading level was fixed at 10 wt% and 1 wt%, respectively by taking as reference previous studies [23–25], GNP loading level was changed as 0.1, 0.3 and 0.5 wt% in the composites. Code names and compounding ratios of composites were given in Table 1.

2.2. Composite characterization

Adhesive wear test was conducted in order to examine the coefficient of friction (COF) of the composites on a pin on-disc tester (Nanovea Tribometer). For this purpose, a ceramic ball with 3 mm

Table 1
Compounding ratios and code names of composites.

PA6,6/CF/GNP/PBO	PA6,6 (wt.%)	CF (wt.%)	GNP (wt.%)	PBO (wt.%)
PA6,6	100	0	0	0
PA6,6_CF	90	10	0	0
0.1GNP	99.9	0	0.1	0
0.3GNP	99.7	0	0.3	0
0.5GNP	99.5	0	0.5	0
CF_0.1GNP	89.9	10	0.1	0
CF_0.3GNP	89.7	10	0.3	0
CF_0.5GNP	89.5	10	0.5	0
CF_0.1GNP_PBO	88.9	10	0.1	1
CF_0.3GNP_PBO	88.7	10	0.3	1
CF_0.5GNP_PBO	88.5	10	0.5	1

radius was used. Friction radius was 4 mm, contact load was 35 N, sliding distance was 70 m and speed of disc was 100 rpm, in all wear tests. COF between the ceramic ball and composite was measured and recorded during the adhesive wear test. After the adhesive wear test, wear traces were examined by using a light microscope and wear volume of the samples was calculated according to a standard test method (ASTM G99-05) by assuming that there was no significant pin wear [30]:

$$V = \frac{\pi \times R \times D^3}{6 \times r} \quad (1)$$

where V is the wear volume (mm³); R is the friction radius (mm); D is the wear trace width (mm); and r is the ball radius (mm).

Tensile test was performed with an Instron 4411 universal testing machine. The specimens had a dumbbell shape with length 45 mm, width 4 mm and thickness 2 mm. The crosshead speed was 5 mm/min and the average values were obtained from at least five measurements.

Thermomechanical properties of the materials were analysed by using a DMA (TA Instruments-Q800) in three-point bending mode. The measurements were performed during 25–150 °C temperature range, 5 °C/min heating rate, 1 Hz constant frequency and 35-μm strain amplitude. Tests were applied to rectangular samples 30 mm length, 4 mm width and 2 mm thickness.

Thermal analysis was performed with a differential scanning calorimeter (DSC, TA Instruments model Q200) in order to determine glass transition temperature (T_g), melting temperature (T_m) and heat of fusion (ΔH_m) values of each sample in a temperature range of 25 °C–300 °C with a ramp of 5 °C/min under a nitrogen atmosphere. The relative degree of crystallinity (X_c (%))_{rel} values of the composites were calculated by using heat of fusion values and following expression [23]:

$$X_c(\%)_{rel} = \frac{\Delta H_m}{(1 - \omega_{f/p}) \times \Delta H_m^0} \times 100 \quad (2)$$

where ΔH_m is the heat of fusion of each sample, ω_{f/p} is the weight fraction of CF or GNP and ΔH_m⁰ is the heat of fusion of the neat PA6,6.

The morphologies of the tensile fracture and adhesive wear surfaces of samples were examined by using SEM analysis. SEM analyses were conducted by using a JEOL JCM-6000 model tabletop scanning electron microscope. The samples to be examined were sputter coated with gold before the examination.

3. Results and discussion

3.1. Adhesive wear test

In this study, tribological properties of PA6,6 matrix composites were examined in three consecutive aspects. The first one of these aspects is the effect of only GNP addition to neat PA6,6; the second one is the effect of GNP addition to CF reinforced PA6,6 matrix composites; and third one is the effect of PBO addition to GNP and CF reinforced hybrid composites.

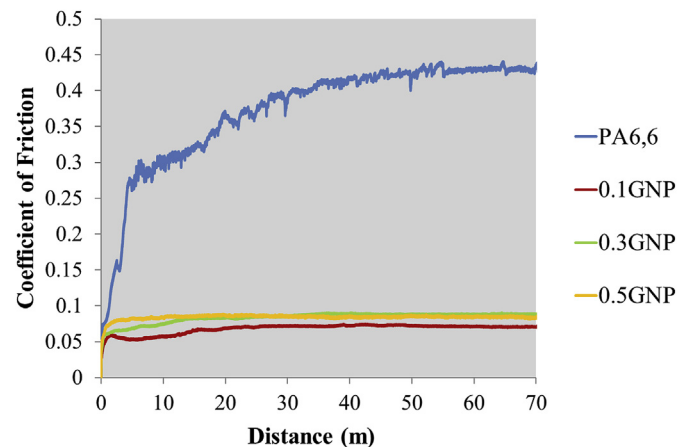


Fig. 1. COF curves of different GNP content filled PA6,6 composites.

Table 2
Wear volume values of samples.

PA6,6/CF/GNP/PBO	Wear volume (mm ³)
PA6,6	0.73 ± 0.1
PA6,6_CF	0.35 ± 0.08
0.1GNP	0.17 ± 0.01
0.3GNP	0.32 ± 0.03
0.5GNP	0.33 ± 0.07
CF_0.1GNP	0.15 ± 0.03
CF_0.3GNP	0.16 ± 0.05
CF_0.5GNP	0.31 ± 0.07
CF_0.1GNP_PBO	0.10 ± 0.06
CF_0.3GNP_PBO	0.12 ± 0.04
CF_0.5GNP_PBO	0.09 ± 0.01

The COF curves of the unfilled and different GNP content filled PA6,6 composites are shown in Fig. 1. Additionally, wear volume values belong to all the samples were given in Table 2. It can be seen from Fig. 1 and Table 2 that all GNP filled PA6,6 matrix composites exhibited much lower COF property and wear volume value than the unfilled PA6,6. As it is known from the literature that weak Van der Waals bonds between the two-dimensional layers of GNPs result in easy inter-layer sliding during wear process. This case causes the arising of lubrication effect of GNP and consequently leads to reduction in COF property [31–34]. In addition, Fig. 1 shows that while 0.1 wt% GNP content filled composites exhibited the lowest COF property among all the composites, 0.3 and 0.5 wt% GNP filled composites exhibited slightly higher COF property than that of 0.1 wt% GNP content filled composites. This slightly increase in COF property with increasing GNP content may result from the agglomeration of GNP at these contents [1,2,7,18–20].

Besides, the effect of only CF and only GNP addition on the COF property and wear volume values of PA6,6 was evaluated, and results were given in Fig. 2 and Table 2. It is seen obviously from Fig. 2 and Table 2 that COF property and wear volume values of PA6,6 was decreased with the addition of CF. This is an expected result because the COF property of polymers can be decreased when they are filled with harder reinforcements such as fibers. These harder reinforcement materials decrease the contact area between adhesive ball and polymer surface during the wear process and thus, a decrement in COF property is observed [35,36]. Fig. 2 also shows that COF property of CF reinforced composites exhibits a similar manner with that of 0.3 and 0.5 wt% GNP filled composites. According to this result, it can be concluded that only 0.1 wt % GNP addition to neat PA6,6 has more remarkable effect on COF property of it than 10 wt% CF addition.

Effect of hybrid CF and GNP reinforcement on the COF property and wear volume value of PA6,6 was investigated and COF curves were

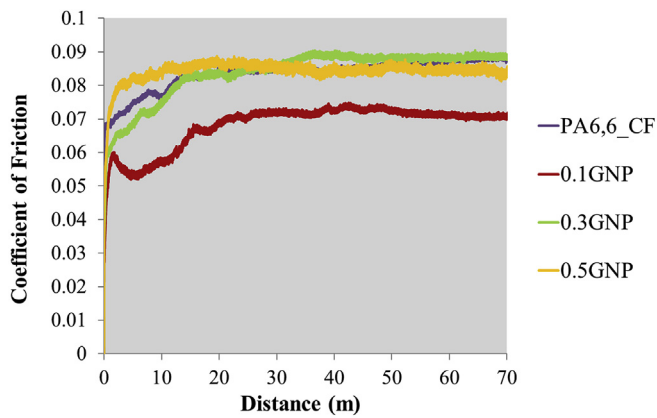


Fig. 2. Comparison the COF values of only CF and only GNP filled PA6,6.

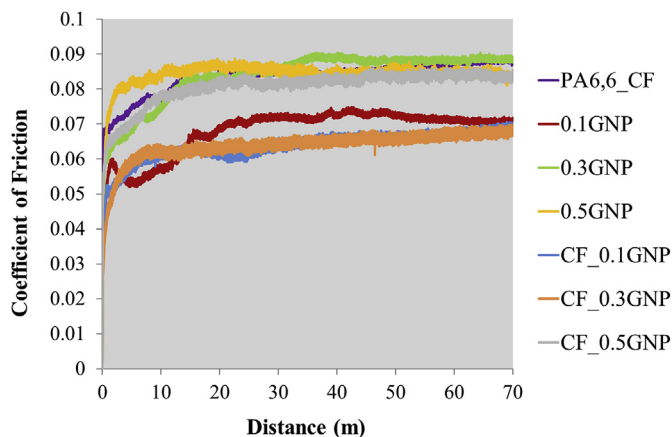


Fig. 3. COF curves of CF and GNP hybrid reinforced PA6,6 composites.

given in Fig. 3. According to Fig. 3 and Table 2, following results can be obtained: (1) 0.1 wt% GNP addition to CF filled composites lowered the COF property and wear volume value, (2) 0.3 wt% GNP addition to CF filled composites lowered the COF property and wear volume value, (3) 0.5 wt% GNP addition to CF filled composites did not considerably improve the COF property and wear volume value of CF filled composites. Consequently, it can be concluded that while hybrid reinforcement makes a positive effect on COF property and wear volume value at lower content of GNP (0.1 and 0.3 wt%), this effect is not possible for higher content of GNP (0.5 wt%). The reason of this hybrid effect on COF property based on the fact that surface functionality of CF and GNP. Because, as mentioned above CF was supplied as treated with a polyamide compatible sizing material and chemical composition of this sizing material includes carbonyl and carboxylic functional groups [25,37]. In addition to this, GNP was supplied as polyamide compatible and it contains oxygen-included functional groups on its surface [38]. Hence, the functional groups are located on GNP can interact with both functional groups on CF surface and PA6,6, so interactions take place between these constituents. These interactions result in homogeneously dispersion and orientation of GN layers in the PA6,6 matrix [39,40]. The orientation of GN layers enables their sliding over each other. Consequently, it can be said that enhanced dispersion and orientation of GN layers in the PA6,6 matrix improve lubricating effect of it and decrease the COF property of relevant composites [32]. However, hybrid effect did not valid for 0.5 wt% GNP filled composites because the agglomeration has strong influence when compared to hybrid effect at this content of GNP.

Effect of coupling agent addition to CF and GNP filled hybrid composites on the COF property and wear volume value of composites was investigated, and COF curves were given in Fig. 4. As it is shown in

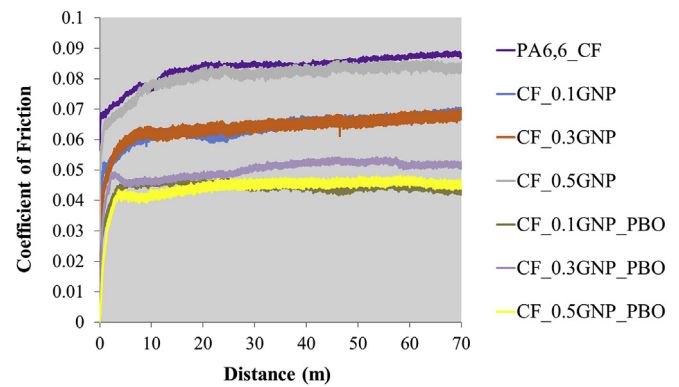


Fig. 4. COF curves of PBO included GNP and CF hybrid reinforced PA6,6 composites.

Fig. 4 and Table 2 that PBO coupling agent addition to CF and GNP filled hybrid composites considerably reduced the COF property and wear volume value. Besides, the highest reduction was obtained for CF_0.5GNP coded sample among all the samples. This result reveals that PBO coupling agent addition to CF and GNP included hybrid composites increased the interaction between constituents due to the mission of PBO in the system. Because, PBO can be used as a chain extender or a coupling agent for polymers which is produced by polycondensation reaction, and PA6,6 is one of these polymers. It performs this mission by coupling amine or carboxylic acid end groups [41]. As a result of this function of PBO, it can be concluded that functional groups of PBO could react with those of CF, GNP and PA6,6. Thus, dispersion and orientation of GNP improves with the addition of PBO and agglomeration disadvantage is overcome especially for 0.5 wt% GNP included composites. Consequently, COF property and wear volume value of composites are reduced with improved lubricating effect of GNP.

3.2. Tensile test

In this study, mechanical properties of PA6,6 matrix composites were examined in three consecutive aspects. The first one of these aspects is the effect of only GNP addition to neat PA6,6; the second one is the effect of GNP addition to CF reinforced PA6,6 matrix composites; and third one is the effect of PBO addition to GNP and CF reinforced hybrid composites.

The tensile strength values belong to unfilled and different GNP content filled PA6,6 matrix composites were given in Fig. 5. According

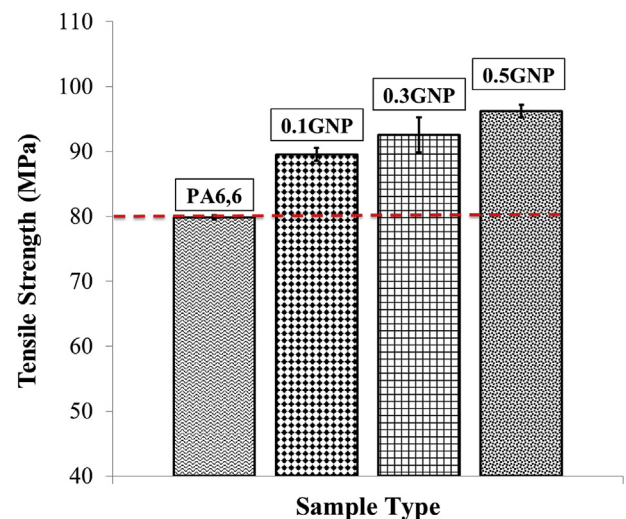


Fig. 5. Tensile strength values of different GNP content filled PA6,6 composites.

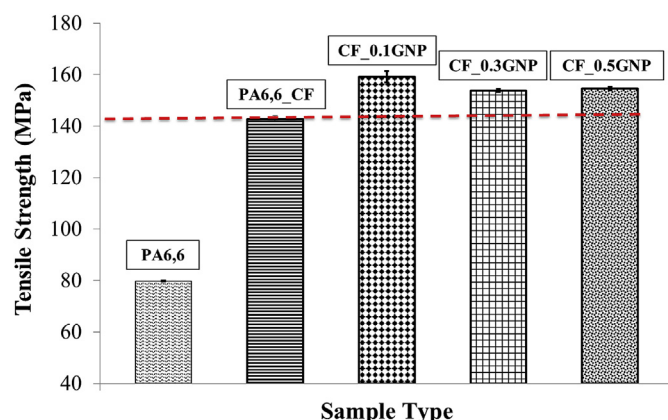


Fig. 6. Tensile strength values of CF and GNP hybrid reinforced PA6,6 composites.

to Fig. 5 it is clear that addition of the GNP leads to a great increase on the tensile strength value of the neat PA6,6. This increase was approximately 12% for 0.1 wt% GNP filled, 15% for 0.3 wt% GNP filled and 20% for 0.5 wt% GNP filled composites. This result may originate from the orientation of basal planes of GNP along the tensile axis. Because when the tensile load applied throughout the direction of basal plane of GNP, covalent C–C bonds along this direction absorb this load. Hence, a significant amount of stress is absorbed with the increasing amounts of oriented GNP along the tensile axis [40]. Correspondingly, it can be said that GNP addition to neat PA6,6 and increasing amount of GNP, increase the tensile strength of neat PA6,6.

The tensile strength values belong to hybrid CF and GNP reinforced PA6,6 matrix composites were assessed and graph belong to these composites were given in Fig. 6. As can be seen from Fig. 6 that GNP addition to CF reinforced PA6,6 matrix composites improves the tensile strength value of it for three loading levels of GNP. This increase was approximately 11% for 0.1 wt% GNP filled hybrid composites and approximately 8% for 0.3 and 0.5 wt% GNP filled hybrid composites. As explained before, the reason of this increase is the interaction between functional groups which are located on GNP, CF and PA6,6 surface. The interactions between these functional groups result in two different strengthening mechanisms. The first one of these mechanisms is the location of GNPs in the CF surface and forming an interphase between CF and PA6,6 [10]. The second one of these mechanisms is more homogeneously dispersion and orientation of GNP in the CF reinforced PA6,6 matrix [39,40]. However, homogeneously dispersion and orientation of GNP in the polymeric matrix is not enough to obtain full of strengthening effect. There is also a need to surrounding of GNP by polymeric matrix. Because, in such a case, a more effective load transfer takes place and reinforcement carries more load. In this regard, Fig. 6 indicates that although all the hybrid composites exhibit improved tensile strength, 0.1 wt% GNP filled hybrid composites exhibits the highest tensile strength value (approximately 11% increase when compared to only CF filled composites). This result can be explained by less agglomeration of GNP at this loading level and correspondingly its higher surface area. Higher surface area of GNP makes more efficiently possible both surrounding of GNP by PA6,6 and load transfer from polymer to GNP.

Effect of coupling agent addition to CF and GNP filled hybrid composites on the tensile strength value was investigated and tensile strength values were given in Fig. 7. Fig. 7 shows that tensile strength values of 0.1 and 0.3 wt% GNP and CF filled hybrid composites decreased with the PBO addition when compared to PBO-free composites. This result may originate from the predominant plasticizer effect of PBO for these compositions of composites [24]. However, tensile strength value of 0.5 wt% GNP and CF filled hybrid composite increased approximately 14% with the PBO addition and this composite exhibited

the best tensile strength value among all composites. The reason of this significant increase may result from the arisen predominant coupling effect of PBO for this composition of composites. As it is known from literature [24], in order to use bisoxazolines as coupling agent, they have to be used in proper amount. If they are used in a lower amount than they should be used, they start to behave as plasticizer and deteriorate the mechanical properties of composites. On the other hand, in the case of they are used in proper amount, the interaction between bisoxazoline, carboxylic end-groups of polyamide and polyamide compatible reinforcement increases. By this way, plasticizer effect of bisoxazoline decreases, and coupling effect of it increases. Hence, it can be said that in this study, interactions between GNP-PA6,6 and CF-PA6,6 are increased with the addition of PBO to GNP and CF included hybrid composites. By this way, interfacial adhesion between GNP-PA6,6 and CF-PA6,6 increased and thus, a more effective load transfer occurred between PA6,6 matrix and reinforcement materials. Additionally, dispersion and orientation of GNP in the composite was further improved with the improved interfacial interaction; and some of well-dispersed GNPs were located in CF surface by forming an interphase between CF and PA6,6; and this led to obtain a better interfacial adhesion between CF and PA6,6 matrix and correspondingly improved tensile strength value [10,24,39].

Load-extension curves, which belong to the best-resulted samples in each group (0.5GNP, CF_0.1GNP and CF_0.5GNP_PBO), were also given in Fig. 8. It can be seen from this figure that CF_0.5GNP_PBO coded hybrid composites exhibit the highest load bearing capacity among other samples.

3.3. Dynamic mechanical analysis

DMA is a thermomechanical analysis method and it can be used to evaluate the interfacial adhesion for fiber or particulate reinforced polymeric matrix composites. Damping factor (tan delta) property that is obtained from DMA can be used for this purpose as well. Because, tan delta is a sensitive indicator for molecular motions which take place in a material. In a fiber-reinforced polymeric matrix composite material, the molecular motions that occur at the interface between fiber and matrix contribute to the damping property of the material separately from the other components. Because in a composite system, which includes fiber, polymeric matrix and fiber-matrix interface, mechanical energy dissipating takes place at polymeric matrix and fiber-matrix interface, with the applied load to material. Therefore, the peak intensity of tan delta is an indicative of the tendency of the material to dissipating mechanical energy and the lower magnitude of tan delta peak indicates the better interfacial adhesion between fiber-matrix and lower mechanical energy dissipation tendency. Consequently, examination of the damping phenomenon helps in evaluating the quality of the interfacial adhesion [25,42,43].

Likewise, in this study, interfacial adhesion quality of composites was evaluated by using DMA. The variation of tan delta of the unfilled and different GNP content filled PA6,6 matrix composites were given in Fig. 9 as a function of temperature. Fig. 9 shows that tan delta peak intensity belongs to neat PA6,6 decreased with the addition of GNP and the minimum peak intensity was obtained for 0.5 wt% GNP included composites. This result coincides with mechanical test results and confirms that increasing amount of GNP decreases tendency of material to dissipating mechanical energy due to the increasing strengthening effect of GNP.

Besides, as it is known that the temperature, which corresponds to the maximum of tan delta peak, gives the glass transition temperature (T_g) of the material. As it is shown from Fig. 9 that GNP addition to PA6,6 matrix shifted the tan delta curves to higher temperatures and correspondingly it can be said that GNP addition increased the T_g value of neat PA6,6. This case can be explained as the immobilization effect of GNP on PA6,6 matrix. Because, according to this effect, GNPs behave as barrier during the viscous flow of polymer chains and they decrease the

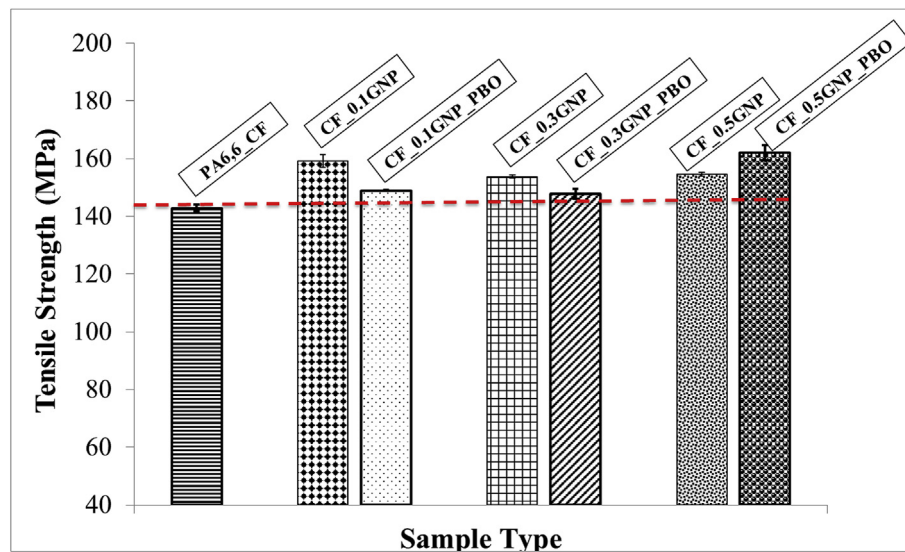


Fig. 7. Tensile strength values of PBO included GNP and CF hybrid reinforced PA6,6 composites.

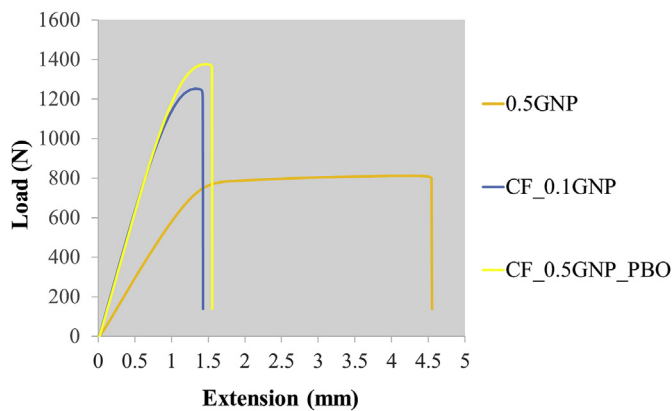


Fig. 8. Load-extension curves of the best-resulted samples in each group of the composites.

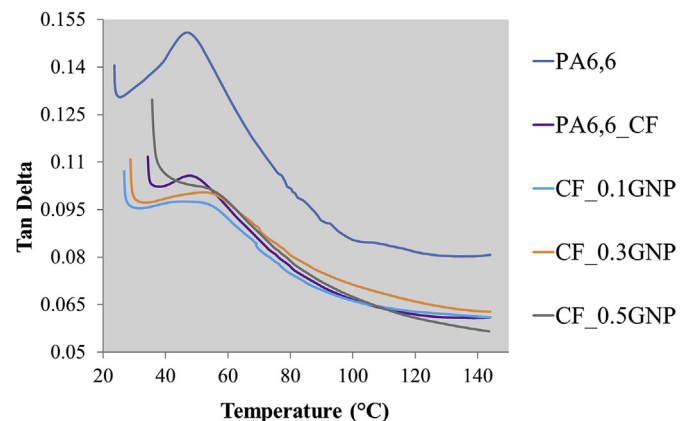


Fig. 10. Tan delta curves of CF and GNP hybrid reinforced PA6,6 composites.

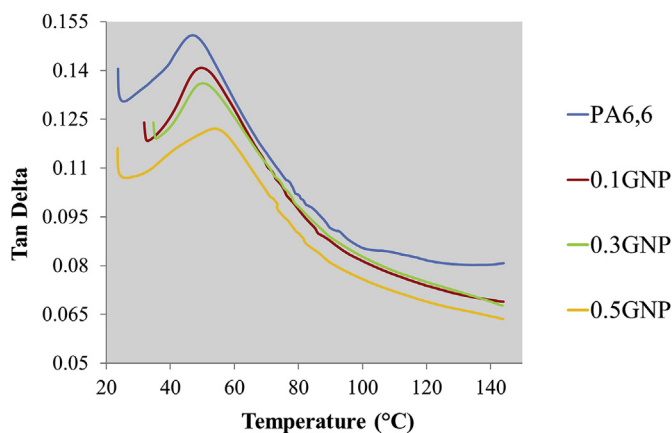


Fig. 9. Tan delta curves of different GNP content filled PA6,6 composites.

mobility of it. This increasement in T_g can be considered because of the effect of homogeneous dispersion and orientation of GNP in to the PA6,6 matrix [43].

The tan delta curves belong to hybrid CF and GNP reinforced PA6,6 matrix composites were given in Fig. 10. It is clearly seen from figure that CF addition to neat PA6,6 decreased the peak intensity and correspondingly dissipating mechanical energy. In addition to this, while

peak intensity of tan delta curves decreased, these curves also shifted to higher temperatures with GNP addition to CF reinforced PA6,6 matrix composites. However, the lowest peak magnitude was observed for 0.1 wt% GNP included hybrid composites and this result is coincided with mechanical test results. According to this result, it can be said that GNP addition increased the T_g value of CF reinforced PA6,6 matrix composites. This increasement in T_g can be considered because of the effect of improved interfacial adhesion between the PA6,6 matrix, CF and GNP and homogeneous dispersion and orientation of GNP in to the CF reinforced PA6,6 matrix.

It can be also pointed out from the Fig. 10 that the tan delta peak broadens with the addition of GNP in to the CF reinforced PA6,6 matrix. The broadening of tan delta peak in the presence of GNP can be ascribed to improved interfacial adhesion between the PA6,6 matrix, CF and GNP. Because, when interfacial interactions take place between filler and matrix, chemical composition of polymer that surrounds the filler (GNP and CF) alters when compared to the bulk matrix. Thereby, this chemically altered part of polymeric matrix inhibits the relaxation of the whole matrix and this case results in a broad tan delta peak [42].

The tan delta curves belong to coupling agent included hybrid CF and GNP reinforced PA6,6 matrix composites were given in Fig. 11. It is obviously seen from figure that peak intensity and correspondingly dissipating mechanical energy of coupling agent and 0.5 wt% GNP included hybrid composites is the lowest when compared to other composites. It can be also pointed out from the figure that tan delta peak shifted to higher temperature with the addition of coupling agent to

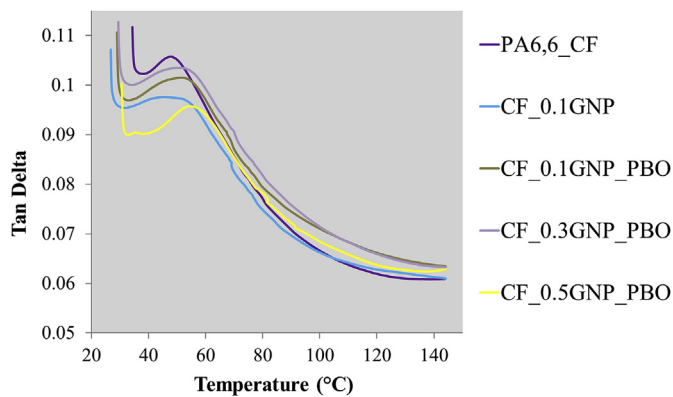


Fig. 11. Tan delta curves of PBO included GNP and CF hybrid reinforced PA6,6 composites.

0.5 wt% GNP included hybrid composites. These results are coincided with mechanical test results and can be ascribed to improved interfacial adhesion between the PA6,6, GNP and CF in the presence of PBO.

3.4. Differential scanning calorimetry analysis

From the DSC thermograms, T_g , T_m and ΔH_m values of each sample were obtained. Besides, $X_c (\%)_{rel}$ values was calculated and results were listed in Table 3. It can be seen from Table 3 that while CF addition to neat PA6,6 slightly increased its T_g value, T_m value of it did not change with the addition of CF. On the other hand, $X_c (\%)_{rel}$ value of neat PA6,6 decreased with the addition of CF. This decrement in $X_c (\%)_{rel}$ value may results from the preventative effect of fibers on the polymer chains, at this loading level of CF. Because, as it is known from literature [23], at low fiber concentrations, crystallinity degree of polymers increases due to the increasing number of nucleation sites. On the other hand, after the reach to critical fiber weight percentage level, mobility of polymer chains starts to decrease with the increasing loading level of fibers and crystal growth in the polymer matrix starts to decrease due to the this preventative effect. As a result, it can be concluded that weight percentage of CFs in the (10 wt%) composites exceeded this critical fiber weight percentage level. For this reason, CFs started to prevent the mobility of PA6,6 chains and prevented the PA6,6 chains from forming crystal structures. Table 3 also shows that, GNP addition to neat PA6,6, GNP addition to CF reinforced PA6,6 matrix composites and PBO addition to GNP and CF reinforced hybrid composites, slightly increased the T_g value of neat PA6,6. However, T_m value of neat PA6,6 did not change with the reinforcement and coupling agent addition. On the other hand, $X_c (\%)_{rel}$ value of neat PA6,6 decreased with the addition of GNP, CF and PBO. In addition to this, CF_0.5GNP_PBO coded sample exhibited the lowest $X_c (\%)_{rel}$ value among all the samples. When it is considered that the this sample exhibited the highest improvement in mechanical, adhesive wear and thermomechanical properties, it can be

Table 3
DSC analysis results of samples.

PA6,6/CF/GNP/PBO	T_g (°C)	T_m (°C)	ΔH_m (J/g)	$X_c (\%)_{rel}$
PA6,6	55.71	265.28	72.69	100.00
PA6,6_CF	57.6	265.99	68.93	94.92
0.1GNP	56.29	264.08	68.86	95.02
0.3GNP	57.46	264.7	68.76	95.07
0.5GNP	58.57	265.2	61.27	93.65
CF_0.1GNP	56.81	265.6	63.96	97.88
CF_0.3GNP	57.31	264.67	62.46	95.79
CF_0.5GNP	56.98	264.55	62.09	95.44
CF_0.1GNP_PBO	56.89	264.18	60.38	93.44
CF_0.3GNP_PBO	57.33	265.49	60.13	93.26
CF_0.5GNP_PBO	57.44	264.91	58.89	91.54

concluded that improved fiber-matrix interfacial adhesion with the addition of PBO, further prevented the mobility of polymer matrix chains and the crystal growth.

3.5. Scanning electron microscopy analysis

Tensile fracture surfaces of composites were investigated by using SEM analysis and SEM micrographs belong to these surfaces were given in Fig. 12 and Fig. 13. By means of SEM micrographs, interfacial adhesion between PA6,6, CF and GNP and also coupling agent addition to composites on interfacial adhesion were evaluated. It is obviously seen from Fig. 12 that there are gaps between CF and PA6,6 matrix, there are no polymeric residue on CF surfaces and they are not wrapped by PA6,6 matrix. On the other hand, Fig. 13 shows tensile fracture surfaces of CF, GNP and coupling agent included hybrid composites. It can be seen from these micrographs that CF surfaces were wrapped by PA6,6 matrix and amount of gaps between fiber and matrix lowered. It can be concluded according to SEM observations that, there is a good adhesion between CF and PA6,6 matrix, in the presence of 0.5 wt% GNP and PBO included composites.

SEM micrographs which belong to worn surfaces of the neat PA6,6 and its composites were given in Fig. 14. Fig. 14 shows that while wear tracks are clearly visible along the sliding direction for neat PA6,6, these tracks are barely visible for 0.1 GNP coded composite. This result implies that the adhesive ball contacts with the GNP in place of the polymeric matrix; by this way, inter-layer sliding in multi-layer GNPs occurs in the 0.1 wt% GNP included composites; and consequently wear resistance property of PA6,6 matrix improves. Fig. 14 also shows that only CF addition as well as simultaneously addition of CF and 0.1 wt% GNP to neat PA6,6 altered the texture of the wear trace belongs to neat PA6,6 and made it rougher and wavelike. This alteration in the texture of wear trace results from the fiber wearing mechanism which occurs from contacts of adhesive ball with the CF in place of the polymer matrix. Besides, it can be seen from this figure that wear trace has been almost invisible for CF_0.5GNP_PBO coded sample. This result implies that PBO addition improves the wear resistance property of hybrid composites due to the better adhesion between CF-PA6,6 and GNP-PA6,6. Because in such a case, reinforcement materials do not peel-off from the PA6,6 matrix easily and adhesive ball contacted with reinforcement materials instead of polymeric matrix. Thus, PA6,6 matrix is supported and protected by CF and GNP.

4. Conclusions

In this study, hybrid composites made of PA6,6 reinforced with GNP and CF were characterized in terms of their adhesive wear, mechanical, thermal and thermomechanical properties and the effect of GNP and PBO coupling agent addition on these properties was determined. Adhesive wear test results revealed that while hybrid reinforcement makes a positive effect on COF property at lower content of GNP, this effect did not valid for 0.5 wt% GNP filled composites because the agglomeration has strong influence when compared to hybrid effect at this content of GNP. PBO addition to hybrid composites considerably reduced the COF property and the highest reduction was obtained for 0.5 wt% GNP filled composites. This case resulted from the increased the interaction between constituents with PBO addition. Thus, dispersion and orientation of GNP improved and agglomeration disadvantage is overcome especially for 0.5 wt% GNP included composites. Tensile test results indicated that GNP addition to CF reinforced PA6,6 matrix composites improved the tensile strength value of it. On the other hand, with the addition PBO, further improvement in tensile strength was obtained for only of 0.5 wt% GNP included hybrid composite. The reason of this case is the predominant coupling effect of PBO for this loading level of GNP, contrary to its plasticizer effect. With this coupling effect, interaction between constituents was enhanced; dispersion and orientation of GNP in the composite was further improved; and this

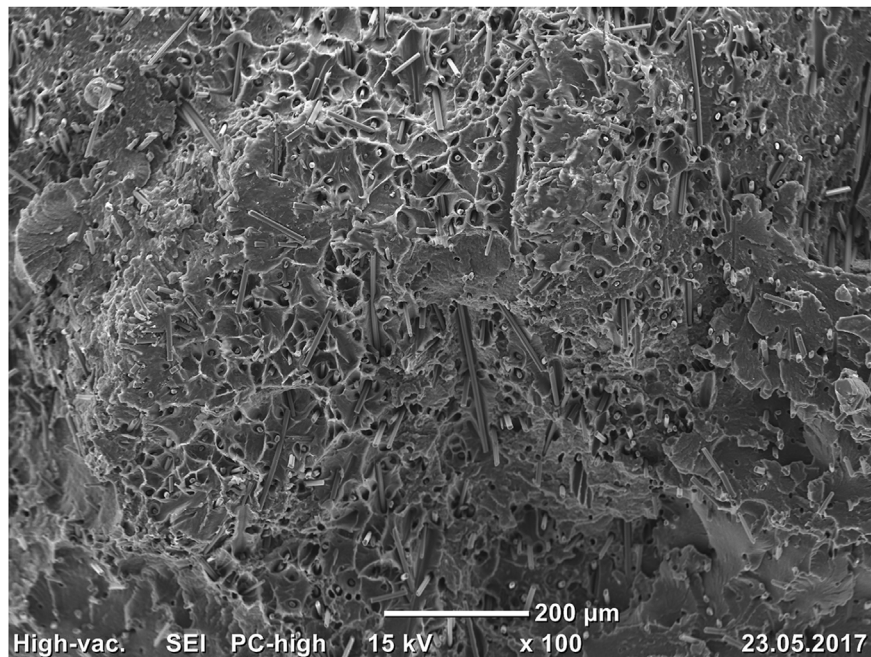


Fig. 12. SEM micrograph belongs to tensile fracture surface of CF reinforced PA6,6 matrix composite.

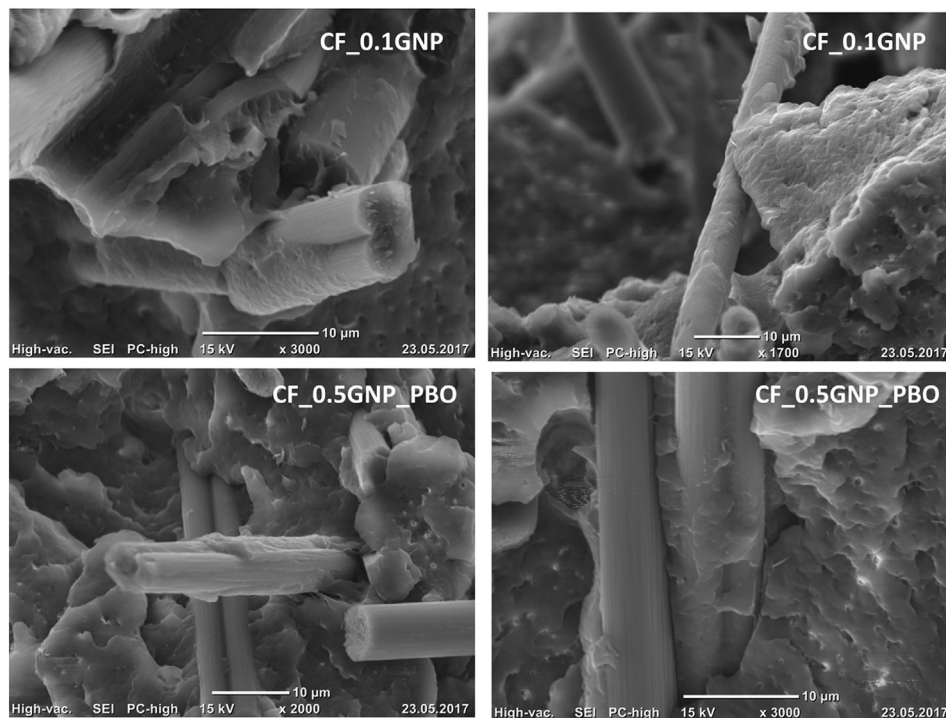


Fig. 13. SEM micrographs belong to tensile fracture surfaces of PBO included hybrid composites.

case reflected to tensile strength values. DMA test results showed that with GNP addition to CF reinforced PA6,6 matrix composites, peak intensity of tan delta curves decreased, these curves shifted to higher temperatures and tan delta peak broadened due to the improved interfacial adhesion between the constituents. These evidences about improved interfacial adhesion between constituents further increased only for the CF_0.5GNP_PBO coded hybrid composites. DSC analysis results revealed that T_g value of neat PA6,6 was slightly increased and $X_c (\%)_{rel}$ value of it decreased with the addition of reinforcement materials and coupling agent. While this decrement in crystallinity was interpreted as the prevention the crystal forming of polymeric matrix

with the addition of reinforcement materials, further decrement in this value was interpreted as prevention the crystal forming of polymeric matrix by improved reinforcement-matrix adhesion. As a results, the highest decrement in $X_c (\%)_{rel}$ value was obtained for CF_0.5GNP_PBO coded sample due to the further improvement in reinforcement-matrix adhesion with the addition of PBO.

Consequently, it can be concluded that GNP addition to CF reinforced PA6,6 matrix composites, resulted in “synergetic effect” or in other words “hybrid effect” and improved tribological, mechanical and thermomechanical properties of composites. However, this effect remained limited only for low loading levels of GNP. In this point, PBO

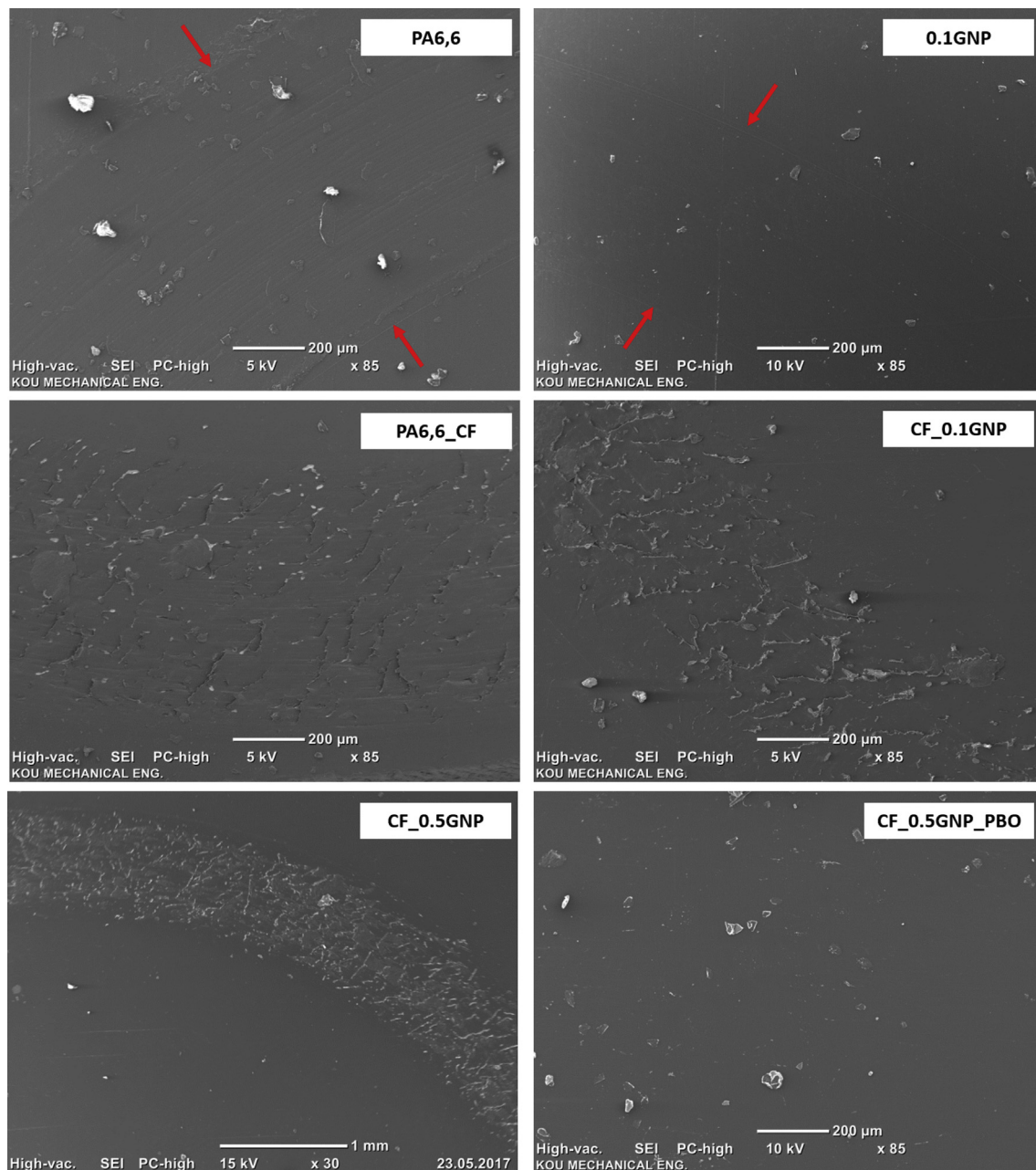


Fig. 14. SEM micrographs belong to adhesive wear surfaces of neat PA6,6 and its composites.

coupling agent which is added to hybrid composites improved the interfacial adhesion between constituents and therefore, CF_0.5GNP_PBO coded hybrid composite exhibited the best ultimate properties among all the samples.

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